Metadata: StormSense Comma Separated Values (CSV) and JSON for 72-Hour Forecasts (fore72)

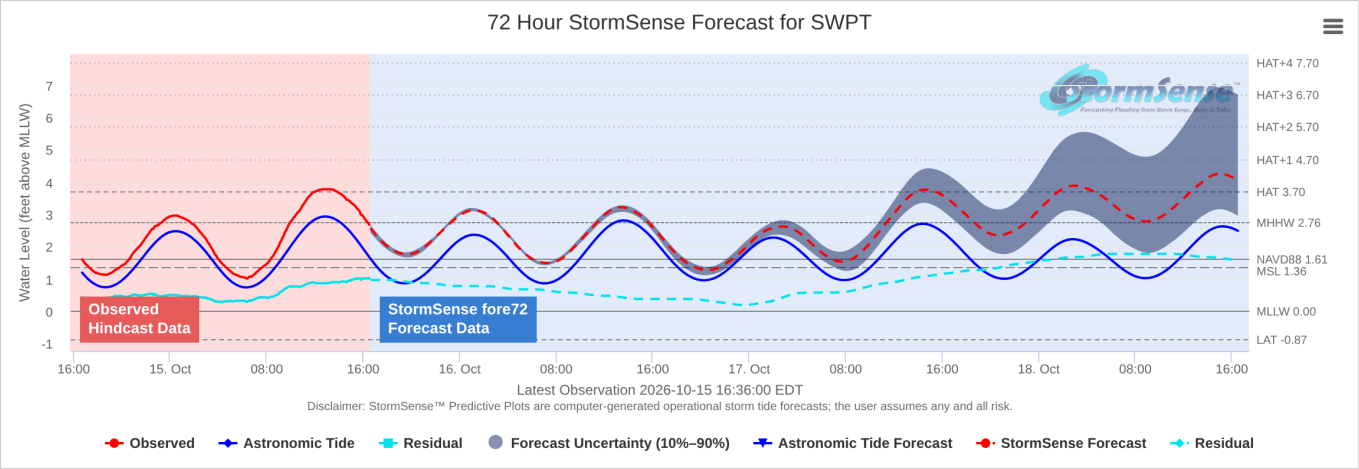


Figure 1. *Example StormSense Chart of 72-hour forecast data (MLLW) at NOAA's Sewells Point tide gauge, in Norfolk, VA, issued October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z) from the P-ETSS 12Z cycle. The red-shaded period is the observed hindcast leading up to the forecast; the blue-shaded period is the 72-hour forecast.*

Figure 1 shows a StormSense 72-hour forecast chart and Table 1 the first rows of the matching download. Each field below is tied to the series it is drawn as in Figure 1 and to its column in Table 1.

Fields **fore72 downloads only include data series for the 72-hour forecast period (the dashed lines and shaded cone in the blue-shaded period of Figure 1; sample rows in Table 1)**

date_time_z	Date and time in Zulu/GMT (UTC), formatted as a string leading with the date as "YYYY-MM-DD" followed by the time (T) as "hh:mm:ss" and Z for Zulu time. Recorded in regular 6-minute intervals (0, 6, 12 … 54 minutes past the hour), listed in chronological order (Table 1, first column).
astro_tide_ft	(**Astronomic Tide Forecast**) Harmonically predicted astronomical tide for the next 72 hours from the NWS P-ETSS model (solid blue line in the forecast period of Figure 1). Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).
surge_ft	Ensemble-mean storm surge from P-ETSS (the Probabilistic Extra-Tropical Storm Surge model of the NWS Meteorological Development Laboratory, run with an ensemble of weather forecasts): the water level change driven by forecast wind and air pressure. Hourly guidance interpolated to 6 minutes and smoothed with a 3-hour low-pass filter. Feet; a height difference, so the same in MLLW and NAVD88.
bias_ft	The observation-based correction P-ETSS adds to its surge, from the recent difference between the gauge's observations and the modeled tide plus surge; it is held or decays through the forecast, and is 0.000 where P-ETSS applies no correction. Feet; the same in MLLW and NAVD88.
residual_ft	(**Residual**) Forecast residual: the non-tidal part of the forecast, SSforecast_ft − astro_tide_ft (surge_ft + bias_ft), drawn as the dashed cyan line in Figure 1. Feet; the same in MLLW and NAVD88.
SSforecast_ft	(**StormSense Forecast**) The StormSense 72-hour storm tide forecast, the ensemble-mean total water level: astro_tide_ft + surge_ft + bias_ft (dashed red line in Figure 1). Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).
SSforecast_p10_ft	(**Forecast Uncertainty (10%–90%)**) 10th percentile of the ensemble total water level (P-ETSS's 90% exceedance level): the lower edge of the shaded cone of uncertainty in Figure 1. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).
SSforecast_p90_ft	Upper edge of the cone in Figure 1: the 90th percentile of the ensemble total water level (P-ETSS's 10% exceedance level). Nine in ten ensemble members fall at or below it. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).
surge_p10_ft, surge_p90_ft	10th and 90th percentiles of the ensemble storm surge (P-ETSS 90% and 10% exceedance). Feet; the same in MLLW and NAVD88.

Table 1. *Example StormSense Table of the first rows of 72-hour forecast data (MLLW) at NOAA's Sewells Point tide gauge, in Norfolk, VA, issued October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z).*

date_time_z	astro_tide_ft	surge_ft	bias_ft	residual_ft	SSforecast_ft	SSforecast_p10_ft	SSforecast_p90_ft	surge_p10_ft	surge_p90_ft
2026-10-15T20:36:00Z	1.610	1.000	-0.052	0.948	2.557	2.417	2.657	0.860	1.100
2026-10-15T20:42:00Z	1.560	1.000	-0.052	0.948	2.507	2.372	2.607	0.864	1.100
2026-10-15T20:48:00Z	1.512	1.000	-0.052	0.948	2.459	2.328	2.559	0.869	1.100
2026-10-15T20:54:00Z	1.465	1.000	-0.052	0.948	2.413	2.286	2.512	0.873	1.100
2026-10-15T21:00:00Z	1.418	1.000	-0.052	0.948	2.367	2.244	2.466	0.877	1.099
2026-10-15T21:06:00Z	1.374	1.000	-0.052	0.948	2.322	2.203	2.421	0.882	1.099
2026-10-15T21:12:00Z	1.331	1.000	-0.051	0.949	2.279	2.164	2.377	0.885	1.098
2026-10-15T21:18:00Z	1.289	1.000	-0.051	0.949	2.237	2.126	2.334	0.889	1.097

P-ETSS levels are tied to each gauge's own record: the median difference between the observations and P-ETSS over the 24 hours before the forecast starts is added to astro_tide_ft, SSforecast_ft and its percentiles (0.00 ft at most NOAA gauges). At stations without a P-ETSS point, the files keep the same fields with the 36-hour forecast in astro_tide_ft, residual_ft and SSforecast_ft and the P-ETSS-only fields left blank (null in the JSON).

Each CSV has a JSON twin at the same address with .json in place of .csv: station, product, datum, units, time_zone, updated, source, fields and a link to this metadata, then a data list with one record per CSV row keyed by the field names above (numbers in feet, blanks as null).

Datums *(Gray reference lines labeled at the right of Figure 1; click a label to hide or show its line.)*

····· **HAT+1 … HAT+4**	Reference lines 1, 2, 3 and 4 feet above HAT, for reading how far an extreme water level rises above the highest predicted tide. Where a station has no published HAT, the steps start from MHHW.
– – – **HAT**	Highest Astronomical Tide: the highest water level predicted from the tide alone (no weather) over the National Tidal Datum Epoch (NTDE, 1983–2001).
- - - - **MHHW**	Mean Higher High Water: the average of the higher of each day's two high tides over the NTDE; a common threshold for nuisance (high-tide) flooding.
—— **NAVD88**	North American Vertical Datum of 1988: the fixed, land-based geodetic datum used for surveyed elevations, roads and flood maps.
– – – **MSL**	Mean Sea Level: the average of hourly water levels over the NTDE.
—— **MLLW**	Mean Lower Low Water: the average of the lower of each day's two low tides over the NTDE. It is zero on these charts and in these downloads.
– – – **LAT**	Lowest Astronomical Tide: the lowest water level predicted from the tide alone over the NTDE.

Datum source: At NOAA-operated tide gauges the datums are NOAA's published tidal datums for that station. USGS, VDEM and StormSense sensors natively report live water levels in NAVD88 only, so their MLLW forecasts and observations are datum-shifted from NAVD88 using the MLLW–NAVD88 separation derived from NOAA VDatum (https://vdatum.noaa.gov/vdatumweb/), and their tidal datum lines come from VDatum as well. Not every datum is available at every station.